CMCP Beyond Linear Score Space

Typed Information-Geometric Evidence Accounting,
Maintenance Credit, Path Auditing, and Falsification-First Evaluation

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Abstract

The Causal Memory and Credit Protocol (CMCP) is intended to let an adaptive system reuse a persistent shared causal mechanism without repeatedly counting correlated manifestations of the same causal innovation as independent evidence. Recent experiments validate an important part of this objective: a persistent typed ledger can prevent duplicate packets from inflating effective precision and can improve same-cohort accuracy and calibration. The same experiments also expose two limitations in a simple implementation. First, a Schur-complement residual computed in a selected linear score representation can mistake a deterministic nonlinear transformation for almost wholly novel evidence. Second, suppressing duplicate evidence can weaken continual-learning retention when replay is serving an optimization-maintenance role in addition to an evidence-assimilation role.

This paper develops a broadened protocol. The central proposal is to replace a single scalar novelty score by a typed geometric signature that separates: (i) independent information increment, (ii) residual predictive or interventional effect, (iii) directional agreement or contradiction, (iv) order dependence and holonomy, and (v) maintenance need. A packet may therefore receive zero shared-memory precision while still receiving nonzero maintenance credit. The information-geometric core uses Fisher-Rao/Hellinger response geometry, tangent vectors, parallel transport, conditional Fisher or Schur residuals, and finite path audits. Distance correlation and t-norm aggregation are retained only as auxiliary dependence alarms and policy gates, not as semantic definitions of redundancy.

The main body presents the conceptual and mathematical framework, gives an implementable staged protocol, and identifies which claims are established, proposed, or still open. A detailed appendix specifies a falsification-first testing program aimed at coding agents, including software interfaces, fixtures, experiment matrices, invariants, acceptance criteria, sealed confirmation procedures, and required artifacts.

Contents

1 Motivation and Scope

1.1 The problem CMCP is trying to solve

Many adaptive systems encounter the same underlying cause repeatedly through different observations, tasks, transformations, agents, or representational pathways. The system should be able to reuse what it has learned about that cause, but it should not automatically treat every manifestation as an independent increment of evidence. The two desiderata are therefore:

1. **read-many reuse:** a shared mechanism can be queried by many contexts;
2. **write-gated assimilation:** the persistent shared record changes only when the incoming packet contains causally novel information or a justified correction.

A credit-only highway complements this memory discipline. High-level error or critique can be routed directly to the modules that could causally reduce it, but the routed message need not become another forward dependency of the model. The ideal division of labor is:

Shared knowledge is widely readable but narrowly writable. Error is globally observable but locally actionable.

The result is not merely a replay filter. It is an attempt at an exactly-once causal assimilation protocol for adaptive systems.

1.2 What the recent experiments changed

The recent evidence-accounting experiments support three conclusions.

1. A typed persistent ledger can prevent repeated manifestations from becoming precision inflation while preserving useful performance.
2. In parameter learning, naive replay can retain old tasks better even while overcounting precision, because replay has a maintenance function that is not identical to new evidence assimilation.
3. A provenance-distinct deterministic transformation can appear almost fully novel to a linear score-space Schur residual. The failure is not the Schur principle itself; it is the chosen representation in which the residual is computed.

These findings motivate a more careful protocol rather than abandonment of the original architecture.

Central design revision. A packet should not receive one undifferentiated "novelty weight." It should receive a typed decision record containing distinct evidence, effect, direction, path, and maintenance quantities. Different downstream actions should consume different fields of that record.

1.3 Broad applicability

The framework is intended to be substrate-neutral. A "packet" may be:

- a batch-aligned neural constraint;
- a gradient, predictive-coding residual, or adjoint;
- an observation or experiment result in a scientific model;
- a temporal-difference error in reinforcement learning;
- an agent-memory claim with provenance;

- a symbolic inference result;
- a world-model transition update;
- a proposed change to a shared expert, adapter, skill, or controller.

Likewise, a "belief state" can be a categorical predictive distribution, a posterior over parameters, a natural-gradient vector, a local Hessian or Fisher operator, a symbolic strength-confidence pair, or another typed state with a meaningful comparison geometry.

2 Experimental Evidence and Its Correct Interpretation

2.1 Persistent precision accounting

In the corrected persistent same-cohort experiment, the CMCPledger produced effective precision near the oracle target while naive accumulation substantially inflated it. The same experiment improved accuracy and calibration under the frozen-weight settlement protocol. This is direct evidence that a typed ledger can enforce the principle

$$\text{repeated causal manifestation} \not\Rightarrow \text{independent precision increment.} \tag{1}$$

The result should be treated as a substantive accounting success. It does not by itself establish universal continual-learning benefit, because the settlement protocol and model state were restricted. It does establish that exactly-once-style assimilation can be operationally meaningful.

2.2 Replay exposed a second control channel

In sequential parameter learning, CMCPkept effective precision near the oracle while naive replay inflated it. Yet naive replay often retained the old task better. This does not refute the accounting objective. Instead it reveals a hidden conflation:

$$\boxed{\text{statistical evidence assimilation} \neq \text{optimization maintenance}} \tag{2}$$

An exact duplicate can deserve zero additional evidence weight while still being useful as a stabilizing training action. Replaying it may prevent later tasks from moving a parameter realization away from the old mechanism. Naive accumulation receives this maintenance benefit by incorrectly pretending the replay is new evidence.

The correct repair is to expose separate controls rather than to restore double counting.

2.3 The bit-flip counterexample

The original score-space criterion used a standardized score vector $s_j \in \mathbb{R}^d$ for a candidate packet j and stored score matrix S:

$$r_{\text{score}}(j \mid S) = \frac{F_{jj} - F_{jS}(F_{SS} + \lambda I)^{\dagger} F_{Sj}}{F_{jj}}, \qquad F_{ab} = d^{-1} s_a^{\top} s_b. \tag{3}$$

This is a normalized Schur-complement residual. It correctly measures the conditional energy remaining outside the stored linear span in the chosen representation. The systematic bit-flip fixture received a value close to one, however, even though it was deterministically related to a stored packet.

The correct diagnosis is:

What failed. The Schur-complement concept did not fail. The score coordinates failed to linearize the relevant equivalence relation. A deterministic nonlinear recoding can be nearly orthogonal in an arbitrary feature representation.

This suggests lifting residualization into a representation defined by packet-induced model responses or local information geometry.

3 Five Quantities That Must Be Kept Separate

A mature CMCPimplementation should classify an incoming packet using at least five conceptually distinct quantities.

3.1 Independent information increment

This asks whether the packet contributes new precision about a mechanism after conditioning on the ledger. Denote it by

$$r_{\text{info}}(j \mid S) \in [0, \infty). \tag{4}$$

An independent repeated measurement can have high r_{info} even when it agrees with the current predictive mean and barely changes the output distribution.

3.2 Residual intervention effect

This asks whether the packet still changes a settled belief after the ledger is active:

$$r_{\text{effect}}(j \mid S) \in [0, \infty). \tag{5}$$

A deterministic descendant may have low residual effect if the source packet has already imposed the same constraint. A corrective packet may have high effect.

3.3 Directional alignment

A scalar magnitude does not distinguish reinforcement from contradiction. Let

$$r_{\text{align}}(j \mid S) \in [-1, 1] \tag{6}$$

measure whether the conditional effect points in the same transported direction as the candidate's standalone effect. Negative values indicate reversal, counterevidence, or a regime mismatch.

3.4 Path dependence

This asks whether assimilation order matters:

$$r_{\text{path}}(j, S) \geq 0. \tag{7}$$

It is a finite measure of whether applying S then j differs from applying j then S. At small update strength it is governed by a commutator; at finite strength it is holonomy-like.

3.5 Maintenance need

This asks whether an already learned mechanism needs stabilizing credit even though the current packet contains no new evidence:

$$r_{\mathrm{maint}}(j \mid S, \theta_t) \geq 0. \tag{8}$$

Maintenance depends on the current learner state, interference pressure, retention loss, and curriculum geometry. It is not a property of the packet alone.

3.6 The resulting typed record

The packet evaluator should emit a record such as

$$\mathcal{R}_{j|S} = (r_{\mathrm{info}}, r_{\mathrm{effect}}, r_{\mathrm{align}}, r_{\mathrm{path}}, r_{\mathrm{maint}}, q_{\mathrm{confidence}}, \texttt{class})\,, \tag{9}$$

where `class` may be one of:

Class	**Interpretation**
redundant	already assimilated, negligible new information and effect
confirmatory	independent information with little predictive displacement
novel	new information and material effect
corrective	effect opposes the current or standalone direction
synergistic	candidate matters more after the ledger than alone
context-local	relevant in one regime but unsafe for shared memory
drift	mechanism appears to have changed over time
branch	one stored mechanism should split into distinct regimes
quarantine	evidence is inconsistent, suspicious, or geometrically unresolved

No single scalar can preserve all of these distinctions.

4 Information-Geometric Response Space

4.1 Why compare packet-induced responses

Raw packet vectors can be misleading because equivalence may be nonlinear in those coordinates. A more intrinsic question is:

> What change does this packet induce in a typed belief state, before and after the existing ledger has been settled?

Fix an ordered cohort C. Let

$$P_0(c) := \text{baseline settled prediction,} \tag{10}$$

$$P_S(c) := \text{prediction after stored ledger } S, \tag{11}$$

$$P_j(c) := \text{prediction after candidate } j \text{ alone,} \tag{12}$$

$$P_{S\cup\{j\}}(c) := \text{prediction after ledger plus candidate.} \tag{13}$$

For categorical outputs, each $P(c)$ lies in a probability simplex.

4.2 Amari alpha divergence and the Hellinger primary

For $-1 < \alpha < 1$, define

$$D_\alpha(P\|Q) = \frac{4}{1-\alpha^2}\left[1 - \sum_k P_k^{(1-\alpha)/2} Q_k^{(1+\alpha)/2}\right]. \tag{14}$$

At $\alpha = 0$,

$$D_0(P,Q) = 4\left(1 - \sum_k \sqrt{P_k Q_k}\right), \tag{15}$$

which is four times the squared Hellinger distance. It is symmetric, bounded on categorical distributions, stable near small probabilities, and locally induces the Fisher-Rao metric.

Define a cohort mean

$$\overline{D}_\alpha(P,Q) = \frac{1}{|C|}\sum_{c\in C} D_\alpha(P(c),Q(c)). \tag{16}$$

A finite conditional influence ratio is

$$r_\alpha^{\text{effect}}(j \mid S) = \frac{\overline{D}_\alpha(P_S, P_{S\cup\{j\}})}{\overline{D}_\alpha(P_0,P_j)+\varepsilon}. \tag{17}$$

For policy use it may later be clipped, but the raw value should be retained during diagnostics. Values above one can indicate synergy or increased sensitivity after prior settlement.

4.3 Local Fisher interpretation

For a small perturbation $P_\epsilon = P + \epsilon\dot{P}$,

$$D_0(P,P_\epsilon) = \frac{\epsilon^2}{2}\sum_k \frac{\dot{P}_k^2}{P_k} + O(\epsilon^3) = \frac{\epsilon^2}{2}\|\dot{P}\|_{F,P}^2 + O(\epsilon^3). \tag{18}$$

Thus the small-strength form of Equation (17) is a conditional-to-marginal Fisher-energy ratio:

$$r_0^{\text{effect}}(j \mid S) \approx \frac{\|\dot{P}_{j|S}\|_{F,P_S}^2}{\|\dot{P}_{j|0}\|_{F,P_0}^2+\varepsilon}. \tag{19}$$

This makes the proposal information-geometric rather than merely heuristic.

4.4 Predictive effect is not information increment

A crucial limitation remains. Two independent measurements may agree exactly and leave the predictive mean unchanged, yet the second measurement can add substantial precision. Therefore

$$r_{\text{effect}} \not\equiv r_{\text{info}}. \tag{20}$$

The following four regimes must be distinguishable:

Residual effect	Conditional information	Interpretation
low	low	duplicate or already assimilated descendant
low	high	independent confirmation; precision increases
high	high	new, corrective, or regime-changing evidence
high	low	recoding, policy constraint, or non-evidential intervention

A robust implementation must therefore compare belief-state displacement and precision increment separately.

5 Conditional Information Accounting

5.1 Conditional score and Fisher information

Suppose packet j has a local score s_j in a statistical model, and stored packet scores are collected into s_S. The conditional Fisher information of j given S is the Schur complement

$$F_{j|S} = F_{jj} - F_{jS}(F_{SS} + \lambda I)^{\dagger} F_{Sj}. \tag{21}$$

A normalized information ratio is

$$r_{\mathrm{info}}(j \mid S) = \frac{\mathrm{Tr}(F_{j|S})}{\mathrm{Tr}(F_{jj}) + \varepsilon}, \tag{22}$$

or a modewise analogue when the information is matrix-valued. Under a suitable generative or posterior model, an exact duplicate should have nearly zero conditional information, while an independent repeated measurement should retain nonzero information.

5.2 Fisher/Hellinger response-space Schur residual

There is a practical bridge between linear score residualization and full finite settlement. Use the square-root embedding

$$\psi(P) = 2\sqrt{P} \tag{23}$$

componentwise, and define each candidate's standalone response signature over the ordered cohort:

$$\phi_j = \mathrm{vec}\left[\psi(P_j(c)) - \psi(P_0(c))\right]_{c \in C}. \tag{24}$$

Construct the response-space Gram matrix

$$K_{ij} = \langle \phi_i, \phi_j \rangle. \tag{25}$$

Then compute

$$r^{\mathrm{FR-Schur}}(j \mid S) = \frac{K_{jj} - K_{jS}(K_{SS} + \lambda I)^{\dagger} K_{Sj}}{K_{jj} + \varepsilon}. \tag{26}$$

This retains exact joint residualization against the full stored set while lifting the linear operation into a nonlinear Fisher/Hellinger response representation. It is cheaper than resettling all packet combinations and more expressive than raw score-space residualization.

5.3 A staged estimator hierarchy

The recommended escalation path is:

1. raw score-space Schur residual;
2. Fisher/Hellinger response-space Schur residual;
3. finite conditional intervention-effect ratio;
4. tangent alignment and path audit when earlier levels disagree.

This avoids paying the most expensive geometric cost for every packet.

6 Tangent Direction, Parallel Transport, and Contradiction

6.1 Why magnitude alone is insufficient

A high residual effect may be consistent evidence, contradiction, synergy, or drift. The direction of the effect matters.

Let

$$u_j = \mathrm{Log}_{P_0}(P_j), \qquad u_{j|S} = \mathrm{Log}_{P_S}(P_{S\cup\{j\}}) \tag{27}$$

be tangent vectors under a chosen information geometry. Transport the standalone vector into the tangent space at P_S:

$$\widetilde{u}_j = \mathrm{PT}_{P_0 \to P_S}\, u_j. \tag{28}$$

Then define

$$r_{\mathrm{mag}} = \frac{\|u_{j|S}\|^2}{\|\widetilde{u}_j\|^2 + \varepsilon}, \tag{29}$$

and

$$r_{\mathrm{align}} = \frac{\langle u_{j|S}, \widetilde{u}_j \rangle}{\|u_{j|S}\|\,\|\widetilde{u}_j\| + \varepsilon}. \tag{30}$$

The pair $(r_{\mathrm{mag}}, r_{\mathrm{align}})$ carries substantially more information than a clipped scalar.

Magnitude	Alignment	Reading
near zero	undefined/irrelevant	effect already absorbed
near one	positive	consistent independent or residual effect
near one	negative	contradiction, reversal, or corrective evidence
above one	positive	synergy or context amplification
moderate	near zero	rotation, mechanism mismatch, or drift

6.2 Square-root simplex geometry

For categorical distributions, the map $P \mapsto \sqrt{P}$ embeds the simplex into the positive orthant of the unit sphere. This gives a tractable approximation to log maps and parallel transport. For implementation, one may use:

- exact spherical log and transport in square-root coordinates;

- local tangent differences with projection onto the simplex tangent plane;
- a small-step Fisher metric approximation when settlements are weak.

The exact spherical formulas are useful for validation even if a cheaper approximation is used online.

7 Order Dependence, Curvature, and Holonomy

7.1 Finite path audit

The conditional effect in Equation (17) examines $S \to j$. It should be complemented by the reverse ordering. Let

$$P_{S\to j} := U_j(U_S(P_0)), \tag{31}$$
$$P_{j\to S} := U_S(U_j(P_0)), \tag{32}$$

where U denotes the relevant settlement or update operator. Define

$$r_{\text{path}}(j, S) = \overline{D}_0(P_{S\to j}, P_{j\to S}). \tag{33}$$

At small strength, if the update generators are H_S and H_j, the leading order difference is governed by

$$[H_S, H_j] = H_S H_j - H_j H_S. \tag{34}$$

At finite strength, Equation (33) is a holonomy-like failure of the accounting frame to close.

7.2 Multiplicity and curvature are different failures

A repeated packet in exactly the same direction may be counted twice while all update operators commute. This is a multiplicity error, not curvature. Conversely, partially shared mechanisms expressed in oblique coordinates can yield noncommutativity and order dependence even when literal duplication is absent.

For rank-one operators

$$H_s = a u u^\top, \qquad H_t = b v v^\top, \qquad \rho = u^\top v, \tag{35}$$

one has

$$\|[H_s, H_t]\|_2 = ab|\rho|\sqrt{1-\rho^2}. \tag{36}$$

Thus exact parallel reuse ($|\rho| = 1$) can overcount without curvature, orthogonal mechanisms ($\rho = 0$) commute safely, and partially re-encoded sharing produces maximal path dependence at intermediate overlap.

7.3 Interference Gram

For multiple contexts define $C_{ij} = [H_i, H_j]$ and

$$G_{(ij),(kl)} = \text{Tr}(C_{ij} C_{kl}). \tag{37}$$

The diagonal measures how strongly each pair conflicts. The off-diagonal terms reveal whether different conflicts occupy the same parameter-space planes. This matters for memory allocation and maintenance routing:

- a nearly diagonal Gram suggests separate interventions or modules;
- a low-rank Gram with large off-diagonal mass suggests one shared correction may service several conflicts.

The Gram is a diagnostic for entangled regimes, not a final proof of modularity. Once the architecture becomes commuting, interference probes become uninformative and must be complemented by convergence, precision, and intervention tests.

8 Dependence Alarms Are Not Redundancy Semantics

8.1 Distance correlation

A useful exploratory quantity is

$$r_{\mathrm{dep}}(j \mid S) = 1 - \max_{i \in S} \mathrm{dCor}(v_j, v_i), \tag{38}$$

where v_j is a raw evidence vector. This can detect nonlinear dependence missed by linear score residualization, including the constructed complement relation in the bit-flip fixture.

But statistical dependence is not equivalent to redundant information. A deterministic relation might be:

1. a known recoding of the same observation;
2. an independently generated but structurally linked observation;
3. counterevidence;
4. a context or regime transformation;
5. corruption or adversarial manipulation.

Pairwise maximum dependence also misses joint dependence that only appears relative to several stored packets.

Recommended role of dependence estimators. Use distance correlation or related tests as suspicion alarms that trigger a richer geometric or semantic audit. Do not use them as the final definition of evidential redundancy.

8.2 Schweizer–Sklar and product-like aggregation

A prototype combined score and evidence novelty using the Schweizer–Sklar t-norm

$$T_p(x, y) = \left[\max\{0, x^p + y^p - 1\}\right]^{1/p}, \qquad p > 0. \tag{39}$$

As $p \to 0^+$, this approaches the product xy. Calibration selected a boundary close to the product limit rather than supporting a distinct non-product geometry.

The conservative interpretation is therefore:

$$r_Q \approx r_{\mathrm{score}}\, r_{\mathrm{dep}} \tag{40}$$

as a transparent policy gate. The t-norm combines decisions already made by component estimators; it does not define causal equivalence or information geometry.

9 Separate Assimilation, Maintenance, and Correction

9.1 Evidence-assimilation weight

Let

$$a_j \in [0, 1] \tag{41}$$

control persistent evidence precision and shared-memory assimilation. For an exact duplicate,

$$a_j \approx 0. \tag{42}$$

For an independent confirmatory measurement, a_j may be positive even if r_{effect} is small.

9.2 Maintenance-credit weight

Let

$$m_j \geq 0 \tag{43}$$

control replay or stabilizing credit. An exact duplicate can legitimately satisfy

$$a_j = 0, \qquad m_j > 0. \tag{44}$$

Maintenance can be driven by:

- declining retention on a stored mechanism;
- high commutator overlap with current tasks;
- displacement of a high-precision mode;
- large parameter drift inside a protected module;
- an explicit replay budget or consolidation schedule.

9.3 Corrective and branch actions

A packet with negative alignment or high path dependence should not be treated as ordinary novelty. It may need a typed action:

$$c_j \in \{\texttt{consistent}, \texttt{corrective}, \texttt{branch}, \texttt{context-local}, \texttt{quarantine}\}. \tag{45}$$

The total learning actuation can be written schematically as

$$\Delta\theta_j = a_j \Delta\theta_{\text{assimilate}} + m_j \Delta\theta_{\text{maintain}} + \Delta\theta_{\text{typed}}(c_j). \tag{46}$$

This equation captures the key lesson of the replay experiments: correct evidence accounting and useful replay pressure are compatible once they are represented by different controls.

10 Interaction with Persistent Causal Memory and a Credit-Only Highway

10.1 Three planes

The combined system can be understood as three planes.

1. **Representation plane:** persistent shared mechanisms plus local, provisional, or context-specific modules.
2. **Credit plane:** mechanism-indexed residuals, geometric transport, responsibility routing, and maintenance credit.
3. **Evidence-ledger plane:** conditional information, duplicate handling, provenance, model-state versioning, promotion, branching, and drift.

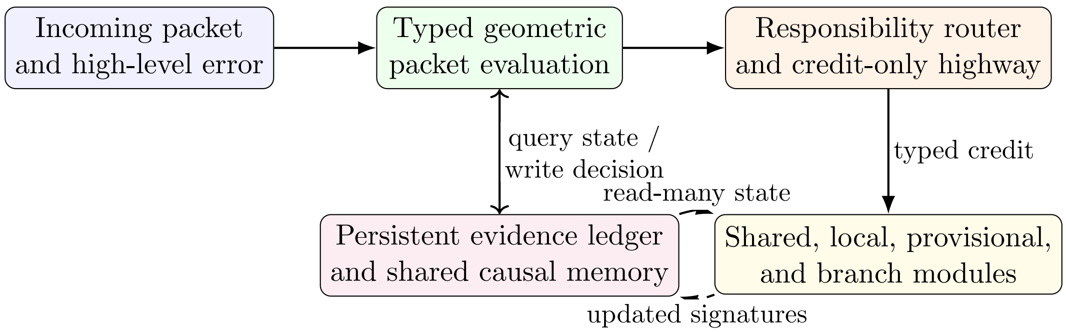

Figure 1: High-level organization of the broadened protocol. Persistent evidence writes, maintenance credit, and forward use are separate actions.

10.2 Credit-only highway update

Let $z_e = C_\phi(e_o, c)$ be a mechanism-indexed error code. A receiving module m may receive

$$\Delta\theta_m = -\eta \sum_j G_{mj,c} T_{m\leftarrow j} \nu_j, \tag{47}$$

where:

- $G_{mj,c}$ estimates responsibility and write permission;
- $T_{m\leftarrow j}$ transports the message into the module's metric and coordinates;
- ν_j is the typed residual after already assimilated information has been removed;
- assimilation and maintenance components of ν_j remain separately labeled.

The highway is credit-only when it changes learning or inference updates without becoming a new dependency of the forward model. This is the safer initial variant because it improves reach without silently changing the represented fixed point.

10.3 Memory promotion and two-stage commit

New or ambiguous information should generally enter a provisional local store first. Promotion to shared memory can require:

$$\texttt{promote} = \mathbf{1}[r_{\text{info}} > \tau_I \wedge q_{\text{responsibility}} > \tau_R \wedge r_{\text{path}} < \tau_P \wedge \texttt{stable-across-contexts}]. \tag{48}$$

If path dependence is high, the packet should be stored in a context adapter or trigger a branch rather than corrupting the stable shared core.

11 Typed Fibers and Model-State Versioning

11.1 The correct comparison space is application-dependent

Settled output distributions are a useful common audit surface, but they may not identify internal mechanisms uniquely. A packet can have little immediate output effect yet carry important local credit. Conversely, different internal updates may produce the same outputs on a narrow cohort.

Define a typed belief or mechanism state B and update operator U_j:

$$B_S = U_S(B_0), \qquad B_{S\cup j} = U_j(B_S). \tag{49}$$

Possible fibers include:

- output probability distributions;
- parameter posteriors;
- module-local predictive distributions;
- natural-gradient or adjoint vectors;
- Fisher, Hessian, or Gauss–Newton operators;
- symbolic belief states;
- value or policy distributions.

A neural CMCP implementation should often maintain both:

1. a global predictive signature for shared-memory assimilation;
2. a module-local credit signature for highway routing and maintenance.

11.2 Versioning

Novelty and redundancy are model-dependent. A packet redundant under checkpoint θ_t may become informative after representation drift. Therefore records should be indexed as

$$\mathcal{R}_{j|S} = \mathcal{R}(j, S; \theta_t, C, v_{\text{settle}}, v_{\text{metric}}). \tag{50}$$

The ledger should store at least:

- cohort and ordered-example identity;
- packet, provenance, and innovation identifiers;
- mechanism family;
- model/checkpoint identity;
- settlement-procedure version;
- metric and estimator version;
- intervention signature;
- precision contribution;
- confidence and revalidation policy;
- values hash and transformation certificates where available.

Cached equivalence decisions should be periodically revalidated on a stable probe bank.

12 A Staged Operational Protocol

12.1 Stage A: hard typed guards

Reject or defer packets under:

- cohort mismatch;
- row-order mismatch;
- duplicate packet identity;
- conflicting payload under one innovation identity;
- model-version incompatibility without a valid transport;
- failed hash or provenance checks.

These guards are exact or fail-closed and should precede statistical estimators.

12.2 Stage B: cheap conditional accounting

Compute raw score-space and response-space Schur residuals. If both indicate clear redundancy or clear independence with high numerical confidence, no expensive settlement is required.

12.3 Stage C: dependence warning

Run distance correlation or a related test. Use it to flag possible nonlinear dependence, not to make the final semantic decision.

12.4 Stage D: intervention geometry

Compute finite Hellinger influence, tangent magnitude, and alignment. Preserve raw values and uncertainty intervals.

12.5 Stage E: path audit

When overlap and effect are substantial, compare both assimilation orders. Use a commutator or finite holonomy proxy to decide whether a shared-memory write is safe.

12.6 Stage F: typed classification

Map the geometric signature to a typed class using transparent rules or a separately validated classifier. The classifier must expose confidence and must be able to defer.

12.7 Stage G: separate actuation

Emit distinct values for:

- evidence assimilation;
- shared-memory write permission;
- maintenance replay;
- corrective credit;
- credit-highway routing;
- provisional storage;
- branch or module allocation.

12.8 Reference pseudocode

Listing 1: High-level typed CMCP evaluation.

```
def evaluate_packet(packet, ledger, model, cohort, config):
    guard = hard_typed_guards(packet, ledger, model, cohort)
    if not guard.ok:
        return Decision.defer(reason=guard.reason)

    score_schur = score_space_schur(packet, ledger, config)
    fr_schur = fisher_response_schur(packet, ledger, model, cohort, config)
    dep_alarm = dependence_alarm(packet, ledger, config)

    need_finite = escalation_policy(score_schur, fr_schur, dep_alarm, config)
    if need_finite:
        effect = conditional_hellinger_effect(packet, ledger, model, cohort)
        tangent = tangent_magnitude_alignment(packet, ledger, model, cohort)
    else:
        effect = approximate_effect(fr_schur)
        tangent = None

    need_path = path_audit_policy(fr_schur, effect, tangent, config)
    path = path_audit(packet, ledger, model, cohort) if need_path else 0.0

    info = conditional_information(packet, ledger, model, cohort, config)
    maint = maintenance_need(packet, ledger, model, config)

    typed_class = classify_typed_record(
        info=info,
        effect=effect,
        alignment=tangent.alignment if tangent else None,
        path=path,
        maintenance=maint,
        provenance=packet.provenance,
    )

    return actuate_separately(
        assimilation_weight=assimilation_policy(info, typed_class),
        maintenance_weight=maintenance_policy(maint, typed_class),
        corrective_action=corrective_policy(typed_class, tangent),
        write_permission=memory_write_policy(typed_class, path),
        routing=credit_route_policy(typed_class, packet, model),
        diagnostics=full_record(...),
    )
```

13 Claim-Status Register

Claim	Status	Evidence or limitation
Persistent typed CMCP can prevent duplicate precision inflation in the tested same-cohort protocol.	Supported	Effective precision remained near the oracle while naive accumulation inflated it; accuracy and calibration also improved in the tested frozen-weight settlement setup.
CMCP universally improves continual-learning retention.	Not established	Naive replay often retained the old task better; replay budget and maintenance pressure were confounded with evidence counting.
A duplicate can deserve zero evidence precision but positive maintenance credit.	Strongly motivated proposal	This directly explains how naive replay can help retention while still overcounting statistical evidence. It requires controlled tests with matched update budgets.
Raw linear score-space Schur residualization is sufficient for arbitrary nonlinear redundancy.	Refuted	A systematic deterministic bit flip was nearly orthogonal in the chosen score representation.
The Schur-complement principle remains useful in a better representation.	Proposed, mathematically motivated	Response-space Fisher/Hellinger signatures preserve joint residualization while representing nonlinear model effects.
Hellinger conditional influence is a full measure of evidence novelty.	Refuted as stated	Independent confirmation can add precision with little predictive displacement. Hellinger influence measures residual effect, not all information increment.
Distance correlation detects some nonlinear dependence missed by score-space residualization.	Supported in the constructed fixture	It caught the systematic complement relation. Its calibration, finite-sample bias, and semantic interpretation remain limited.
A specifically non-product Schweizer–Sklar geometry is supported.	Not supported	Calibration selected the tested boundary close to the product limit.
Path dependence should be measured separately from multiplicity.	Supported by theory	Exact duplicates can overcount while commuting; partially re-encoded shared causes can generate nonzero commutators and holonomy.
The five-field typed record is sufficient for general causal identity.	Open	It is a disciplined decomposition, not a proof of universal causal discovery. Interventions, provenance, and application-specific semantics remain necessary.

14 Limitations and Open Problems

14.1 Causal meaning cannot be recovered from dependence alone

No statistical dependence measure can determine by itself whether two packets are redundant observations, independent measurements, counterevidence, or different regimes. Typed provenance, transformation certificates, interventions, or a credible causal model remain necessary.

14.2 Information geometry is local unless finite settlement is used

Fisher and Hessian quantities are local second-order objects. Strongly nonlinear updates may agree to second order and diverge at larger strength. The staged protocol therefore uses local geometry for screening and finite settled effects for escalation.

14.3 Output geometry may hide internal distinctions

Two mechanisms can induce identical predictions on a narrow probe cohort. Probe banks should be broad, versioned, and mechanism-sensitive. Module-local signatures may be required.

14.4 A low path signal is not a proof of good modularity

A broken or inactive learning path can also yield low interference. Path diagnostics must be paired with effectiveness, convergence, precision, and intervention measurements.

14.5 Maintenance can become a new route to overfitting

Separating maintenance from evidence assimilation prevents precision inflation, but an unconstrained maintenance channel can still monopolize optimization budget. It should be explicitly budgeted and evaluated at matched update norms.

15 Conclusion

Recent experiments improve the CMCP program by exposing two false identifications:

$$\text{linear conditional novelty} \neq \text{general causal redundancy}, \tag{51}$$

and

$$\text{new evidence weight} \neq \text{maintenance replay weight}. \tag{52}$$

The appropriate response is a typed information-geometric protocol rather than a new single scalar. Conditional Fisher information measures independent precision; Hellinger/Fisher response geometry measures residual effect; tangent alignment distinguishes reinforcement from contradiction; commutators and finite path audits measure order dependence; and a separate maintenance channel preserves useful replay without corrupting the evidence ledger.

The resulting design fits naturally with persistent shared causal memory and a causal credit-only highway. Shared mechanisms can be represented once, read broadly, modified only by causally justified innovations, and stabilized by maintenance credit that is explicitly not counted as new evidence. The testing program in the appendix is intended to determine which parts of this architecture survive adversarial fixtures and sequential parameter learning.

A Coding-Agent Testing Plan

A.1 Purpose and operating principles

This appendix specifies a falsification-first implementation and evaluation program. Coding agents should treat every estimator as provisional until it passes held-out mechanism-family tests. The objective is not to tune one fixture set until it looks correct. It is to determine which distinctions are recoverable under explicit contracts.

The program follows six rules:

1. Preserve exact typed guards and fail closed on identity mismatches.
2. Keep assimilation, maintenance, and correction outputs separate in code and logs.
3. Preserve raw, unclipped geometric quantities for analysis.
4. Use held-out transformation families, not only fresh random seeds.
5. Match optimization budgets when comparing replay methods.
6. Seal confirmation seeds and open them only after thresholds are frozen.

A.2 Recommended repository layout

```
cmcp/
  cmcp/
    types.py
    ledger.py
    guards.py
    settlement.py
    geometry/
      alpha_divergence.py
      hellinger.py
      simplex_transport.py
      score_schur.py
      response_schur.py
      conditional_information.py
      path_audit.py
      commutator_probes.py
    dependence/
      distance_correlation.py
      baselines.py
    policy/
      classifier.py
      assimilation.py
      maintenance.py
      correction.py
      routing.py
    diagnostics/
      records.py
      calibration.py
      plots.py
  fixtures/
    generators.py
    transformations.py
    semantics.py
    manifests/
  experiments/
```

```
    configs/
    runners/
    sealed/
  tests/
    unit/
    property/
    integration/
    regression/
  reports/
  scripts/
```

A.3 Core data types

Coding agents should define explicit immutable types. Suggested fields follow.

```
Packet:
  packet_id
  innovation_id
  provenance_id
  mechanism_family
  cohort_id
  ordered_example_ids
  values
  precision
  transformation_certificate
  created_at
  model_version
  settlement_version

LedgerEntry:
  canonical_innovation_id
  contributing_packet_ids
  provenance_graph
  mechanism_family
  accumulated_precision
  intervention_signature
  information_signature
  model_version
  metric_version
  confidence
  last_validated_at

TypedDecision:
  info_ratio
  effect_ratio_raw
  effect_ratio_clipped
  alignment
  path_distance
  dependence_alarm
  maintenance_need
  typed_class
  assimilation_weight
  maintenance_weight
  corrective_action
  shared_write_permission
```

```
route_targets
uncertainty_intervals
guard_status
estimator_versions
```

No downstream module should infer maintenance weight from assimilation weight or vice versa.

A.4 Phase 0: reproducibility and guard audit

A.4.1 Implementation tasks

1. Reproduce the persistent same-cohort ledger result from a clean checkout.
2. Reproduce the parameter-learning replay sweep.
3. Reproduce the provenance-distinct bit-flip failure of raw score-space Schur residualization.
4. Freeze canonical serialization and hashing of ordered cohorts and packet values.
5. Add explicit model, settlement, and metric versions to every ledger entry.

A.4.2 Required guard tests

Each test must fail closed:

- row permutation under the same cohort identifier;
- cohort mismatch;
- duplicate packet identifier;
- same innovation identifier with conflicting payload;
- stale model version without approved transport;
- altered values with unchanged hash field;
- changed settlement code version under a cached decision.

A.4.3 Acceptance criteria

- Exact replays reproduce reported metrics within predefined numerical tolerance.
- Every guard test returns a typed defer/reject result, not an implicit zero weight.
- All experiment manifests include code revision, environment, random seeds, hardware, and configuration hashes.

A.5 Phase 1: divergence and simplex-geometry unit tests

A.5.1 Alpha-divergence tests

Implement Equation (14) in stable log space. Test:

- non-negativity within floating-point tolerance;
- identity of indiscernibles;
- symmetry at $\alpha = 0$;
- convergence to forward and reverse KL at endpoints;
- finite behavior with probabilities near zero using explicit epsilon policy;
- invariance under class permutation;
- cohort-order invariance when ordered identities are permuted consistently.

A.5.2 Square-root geometry tests

Implement exact or validated spherical log maps and parallel transport. Verify:

- norm preservation under transport;
- transported vectors remain tangent to the destination sphere;
- round-trip transport error on small loops;
- agreement of squared tangent norm with Hellinger distance at small step size;
- stable behavior near simplex boundaries.

A.5.3 Gradient checks

Use finite differences or automatic differentiation to validate gradients of:

- Hellinger divergence;
- alpha divergence for several α values;
- tangent magnitude and alignment;
- response-space Gram construction.

A.6 Phase 2: fixture generator with explicit causal semantics

Every generated fixture must include both a mathematical transformation and a semantic label. The same mathematical map can have different semantics in different experimental worlds.

A.6.1 Required fixture families

Fixture	**Construction**	**Expected semantic class**
Exact duplicate	identical packet, same innovation lineage	redundant
Deterministic descendant	known derived packet with certified lineage	redundant or low assimilation
Independent repeat	independent noise realization with same expected value	confirmatory
Invertible recoding	permutation, rotation, monotone or invertible nonlinear map	same mechanism after transport
Systematic complement	bit flip or sign inversion	ambiguous: recoding or corrective, depending on certificate
Noisy copy	controlled signal-to-noise ratio	partial redundancy
Partial overlap	mixture of stored and novel factors	mixed information
Joint redundancy	candidate determined by two stored packets but weak pairwise dependence	redundant only conditionally on the set
Synergy	candidate has little standalone effect but large conditional effect	synergistic
Counterevidence	independent observation opposing current belief	corrective

Fixture	Construction	Expected semantic class
Context split	same observable relation under distinct latent regimes	branch or context-local
Mechanism drift	same identifier but time-varying data-generating rule	drift
Adversarial correlated packet	high dependence but distinct causal source	informative despite dependence

A.6.2 Held-out transformation families

Partition transformations by family, not seed. Example split:

- calibration: affine maps, bit flips, coordinate permutations, Gaussian noisy copies;
- confirmation: nonlinear invertible maps, many-to-one maps, cross-factor maps, context-dependent transforms;
- adversarial: dependence-preserving but semantically informative constructions not seen during calibration.

No learned policy or threshold may see confirmation-family labels before the protocol is frozen.

A.7 Phase 3: estimator comparison

A.7.1 Estimators

Compare:

1. provenance-only deduplication;
2. raw score-space Schur;
3. product of score Schur and distance-correlation clearance;
4. Fisher/Hellinger response-space Schur;
5. finite Hellinger conditional effect;
6. tangent magnitude plus alignment;
7. full typed protocol with path audit;
8. optional learned classifier, only after geometric baselines are stable.

A.7.2 Metrics

Report:

- class-wise precision, recall, and confusion matrix for semantic classes;
- continuous calibration of assimilation weight against oracle conditional information;
- duplicate write amplification: shared-memory commits per true innovation;
- false suppression rate on independent confirmations;
- failure to quarantine counterevidence;
- branch precision and recall;
- compute cost and number of settlement calls;
- robustness across cohort size and probability sharpness.

A.7.3 Primary hypotheses

1. Response-space Schur should outperform raw score-space Schur on nonlinear recodings while retaining exact duplicate behavior.
2. Finite Hellinger influence should detect residual model effect but should not be expected to identify independent confirmation by itself.
3. Alignment should separate consistent novelty from corrective evidence better than magnitude alone.
4. Path auditing should identify intermediate-overlap re-encoding cases that multiplicity estimators miss.
5. Pairwise distance correlation should fail some jointly redundant fixtures, motivating full-set conditioning.

A.8 Phase 4: assimilation versus maintenance experiment

A.8.1 Goal

Determine whether naive replay's retention advantage is due to maintenance pressure rather than correct evidence accounting.

A.8.2 Conditions

Compare at least:

1. no replay;
2. naive replay counted as new evidence;
3. CMCP assimilation only;
4. CMCP with separate maintenance channel;
5. CMCP with maintenance controlled by retention drop;
6. CMCP with maintenance controlled by commutator/interference estimates.

A.8.3 Budget matching

Run two analyses:

1. equal raw replay coefficient;
2. equal total parameter-update norm or equal optimizer-step budget.

Without the second analysis, a method can appear superior merely because it receives more optimization energy.

A.8.4 Measurements

Record:

- old-task retention;
- new-task learning speed and final performance;
- effective evidence precision versus oracle;
- calibration and Brier score;
- cumulative update norm by channel;
- parameter displacement in stored mechanism subspaces;

- number of shared-memory writes;
- maintenance credits issued;
- task-order variance.

A.8.5 Decisive prediction

A successful separated design should match or exceed naive replay retention while keeping the evidence ledger near the oracle precision. If it cannot do so under matched update budget, the maintenance hypothesis is weakened.

A.9 Phase 5: path and curvature experiments

A.9.1 Rotated-overlap fixture

Create two tasks that act on rank-one or low-rank directions with controlled overlap ρ. Sweep $\rho \in [0, 1]$. Measure:

- multiplicity error;
- $\|[H_s, H_t]\|$ or an HVP estimate;
- finite order distance $D_0(P_{s\to t}, P_{t\to s})$;
- task-order performance difference;
- typed classification and memory action.

The expected commutator profile follows Equation (36): zero at orthogonal and parallel extremes, maximal at intermediate overlap.

A.9.2 Interference Gram experiment

For four or more tasks, construct cases in which:

- conflict planes are mutually orthogonal;
- several task pairs share one conflict plane;
- conflict is full-rank and diffuse.

Estimate the Gram in Equation (37) using Hessian-vector or finite-difference probes. Test whether the resulting low-rank structure correctly predicts whether one maintenance buffer, projection, or highway can service several task pairs.

A.10 Phase 6: persistent memory operations

Implement and test:

1. **create:** allocate a provisional mechanism entry;
2. **promote:** move stable cross-context evidence into shared memory;
3. **merge:** combine entries with compatible information and transport signatures;
4. **split:** divide an entry when negative alignment or multimodality persists;
5. **branch:** retain a shared parent with context-specific children;
6. **retire:** mark obsolete mechanisms while preserving provenance;
7. **revalidate:** recompute cached equivalence after model drift.

Every operation must be reversible in the metadata layer and must preserve append-only evidence provenance.

A.11 Phase 7: integration with a credit-only highway

A.11.1 Router evaluation

In synthetic environments with known intervention structure, measure:

- router precision and recall for responsible modules;
- amount of credit delivered to unrelated modules;
- retention and transfer;
- shared-memory write precision;
- interference Gram before and after routing;
- fixed-point drift relative to a no-highway baseline.

A.11.2 Five architecture conditions

Compare:

1. local learning without shared memory or highway;
2. shared memory without a highway;
3. raw full-error highway without memory;
4. causally routed highway without persistent evidence accounting;
5. full shared memory plus typed causal credit-only highway.

The central prediction is that the raw highway improves reach but broadens interference, while the combined system preserves reach and concentrates interference into explicit shared subspaces.

A.12 Phase 8: sealed confirmation

Before opening confirmation seeds or held-out transformation families, freeze:

- estimator formulas;
- numerical stabilizers;
- thresholds;
- escalation policy;
- semantic classifier rules;
- acceptance criteria;
- primary and secondary metrics.

The sealed report must include all failed conditions and not only the best configuration.

A.13 Acceptance gates

A suggested minimum set of gates follows.

Gate	**Criterion**
Accounting	Effective precision remains within a preregistered tolerance of oracle under duplicate bursts.
Independent confirmation	False suppression of independent repeated measurements remains below threshold.

Gate	Criterion
Nonlinear redundancy	Response-space or finite geometry substantially reduces the bit-flip/invertible-recoding failure relative to raw score Schur.
Correction	Negative-alignment fixtures are not silently suppressed as duplicates.
Joint dependence	Full-set residualization outperforms pairwise maximum dependence on jointly redundant fixtures.
Maintenance	Separate maintenance matches or exceeds naive replay retention under equal update budget without precision inflation.
Path detection	Rotated-overlap order dependence follows the predicted intermediate-overlap profile.
Memory branching	Context-split and drift fixtures trigger branches or local storage with acceptable precision and recall.
Highway localization	Responsible-module recall improves without broad unrelated-module credit or diffuse commutator growth.
Reproducibility	Independent rerun from manifest reproduces all primary metrics within tolerance.

A.14 Required experiment artifacts

Every run should emit:

- a machine-readable manifest;
- exact code revision and dirty-state flag;
- environment and dependency lock;
- hardware description;
- seed list and sealed/unsealed status;
- fixture semantic manifest;
- raw per-packet typed records;
- aggregate metrics with uncertainty intervals;
- update-norm accounting by channel;
- plots of information, effect, alignment, path, and maintenance quantities;
- failure cases and representative packet traces;
- a concise human-readable run report.

A.15 Implementation order for coding agents

1. Reproduce and lock the existing ledger experiments.
2. Implement stable Hellinger/alpha divergence and unit tests.
3. Implement response-space signatures and FR-Schur residualization.
4. Add tangent magnitude and alignment.
5. Add separate assimilation and maintenance outputs before changing policy behavior.
6. Run fixture-only estimator comparisons.
7. Implement finite path audit and commutator probes.
8. Run matched-budget sequential learning.

9. Add memory branch/promotion operations.
10. Integrate the credit-only highway last, after the accounting layer is independently validated.

A.16 Stop conditions

Coding agents should stop and report rather than silently patch when:

- the same semantic fixture receives unstable labels across small numerical changes;
- a divergence result depends strongly on arbitrary epsilon choice;
- confirmation-family performance collapses relative to calibration fixtures;
- maintenance gains disappear under matched update budget;
- path distance is low only because updates are ineffective;
- a learned estimator beats baselines only on seen transformation grammars;
- model-version drift invalidates cached packet signatures.

B Mathematical Notes for Implementers

B.1 Stable Hellinger computation

For normalized categorical vectors p, q with floor δ applied only for log-based endpoint calculations, compute Hellinger directly as

$$D_0(p,q) = 4\left(1 - \sum_k \sqrt{p_k q_k}\right). \tag{53}$$

Clip the affinity to $[0,1]$ only to remove floating-point excursions, not as a semantic operation.

B.2 Hutchinson estimate for commutator norms

To estimate

$$\|[H_i, H_j]\|_F^2 = \mathrm{Tr}([H_i, H_j]^\top [H_i, H_j]), \tag{54}$$

sample Rademacher vectors z_r and compute

$$\|\widehat{[H_i, H_j]}\|_F^2 = \frac{1}{R}\sum_{r=1}^{R} \|H_i(H_j z_r) - H_j(H_i z_r)\|_2^2. \tag{55}$$

No dense Hessian is required if Hessian-vector products are available.

B.3 Confidence intervals

For Monte Carlo settled predictions, use common random numbers across $P_0, P_S, P_j, P_{S\cup j}$ to reduce estimator variance. Report bootstrap or repeated-seed intervals for all ratios. Avoid treating a noisy denominator near zero as meaningful novelty; classify it as low-standalone-effect and defer to information/provenance logic.

B.4 Numerical rank and regularization

For Schur complements:

- report the spectrum or effective rank of F_{SS} or K_{SS};
- sweep λ on calibration data only;
- use a pseudoinverse tolerance tied to matrix scale;
- log the residual before and after clipping;
- test invariance under orthogonal coordinate transforms.

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